

SECTION A

Attempt All Questions

Multiple Choice Questions

[30 x 1 = 30]

1. What is the key technology that enables ultra-low latency in 5G?
 - a) OFDMA
 - b) **Network Slicing**
 - c) MIMO
 - d) Beamforming
2. What does "MIMO" stand for?
 - a) Multiple Input Mobile Output
 - b) **Multiple Input Multiple Output**
 - c) Mobile Internet Mobile Output
 - d) Multi-Interface Mobile Operation
3. Which of the following is a key component of IoT?
 - a) **Sensors**
 - b) Paper documents
 - c) Manual switches
 - d) Landline telephones
4. What is the primary purpose of IoT?
 - a) To replace human workers
 - b) **To connect and exchange data between devices over the internet**
 - c) To increase energy consumption
 - d) To reduce internet usage
5. What is the "collision domain" in CSMA/CD?
 - a) A secure area of the network
 - b) **A segment where collisions can occur between competing devices**
 - c) A protocol for error correction
 - d) A method for encrypting data
6. How does CSMA/CA handle potential collisions differently from CSMA/CD?
 - a) It ignores collisions completely
 - b) **It uses techniques like RTS/CTS (Request-to-Send/Clear-to-Send) to reserve the channel**
 - c) It stops transmission after detecting a collision
 - d) It only works in wired networks
7. What is the primary purpose of signaling in telecommunication networks?
 - a) To increase data transmission speed
 - b) **To establish, manage, and terminate communication sessions**
 - c) To encrypt user data
 - d) To provide power to network devices

8. Which signaling system is widely used in traditional PSTN (Public Switched Telephone Network)?
 - a) SIP (Session Initiation Protocol)
 - b) **SS7 (Signaling System No. 7)**
 - c) TCP/IP
 - d) HTTP
9. What does ISDN stand for?
 - a) International Services Digital Network
 - b) **Integrated Services Digital Network**
 - c) Internet Service Data Node
 - d) International Subscriber Dialing Number
10. Which ISDN channel is used for control and signaling purposes?
 - a) B-channel (Bearer channel)
 - b) **D-channel (Delta channel)**
 - c) E-channel (Echo channel)
 - d) H-channel (High-speed channel)
11. What is the data rate of a basic ISDN BRI (Basic Rate Interface)?
 - a) 1.544 Mbps
 - b) **144 Kbps**
 - c) 2.048 Mbps (E1 speed)
 - d) 10 Mbps (Ethernet speed)
12. Which generation of mobile networks introduced digital voice communication?
 - a) 1G
 - b) **2G**
 - c) 3G
 - d) 4G
13. What is the typical frequency band for LTE (4G) in most countries?
 - a) 900 MHz
 - b) **1800 MHz (Band 3)**
 - c) 2.4 GHz
 - d) 60 GHz
14. What is the frequency range of GSM-900 used in 2G networks?
 - a) 1800-1900 MHz
 - b) **890-960 MHz**
 - c) 2.4-2.5 GHz
 - d) 800-850 MHz
15. What is the basic structural and functional unit of a cellular network?
 - a) **Cell**
 - b) Base Transceiver Station (BTS)

- c) Mobile Switching Center (MSC)
 - d) Frequency Band
16. Which technology allows frequency reuse in cellular systems?
- a) TDMA (Time Division Multiple Access)
 - b) **Spatial separation of cells using the same frequency**
 - c) Increasing transmission power
 - d) Using only licensed spectrum
17. What is the main purpose of handoff (handover) in cellular networks?
- a) To increase signal bandwidth
 - b) **To maintain connectivity when a user moves between cells**
 - c) To encrypt voice calls
 - d) To reduce call setup time
18. Which of the following is a key objective of cellular network optimization?
- a) Increasing cell size to reduce infrastructure cost
 - b) **Improving Quality of Service (QoS) and network efficiency**
 - c) Using only one frequency band for all cells
 - d) Eliminating handover procedures
19. What is the primary purpose of frequency reuse in cellular optimization?
- a) To assign unique frequencies to every cell
 - b) **To maximize spectral efficiency by reusing frequencies in non-adjacent cells**
 - c) To reduce the number of base stations
 - d) To increase transmission power
20. Which optimization technique helps reduce co-channel interference?
- a) Increasing cell size
 - b) **Adjusting cell cluster size and sectoring**
 - c) Using only omni-directional antennas
 - d) Disabling handovers
21. What does drive testing primarily help optimize in a cellular network?
- a) Core network latency
 - b) **Real-world coverage, signal strength, and handover performance**
 - c) Battery life of mobile devices
 - d) Internet backbone speed
22. Which parameter is adjusted during cell breathing in CDMA networks?
- a) **Cell coverage area dynamically shrinks/expands based on user load**
 - b) Fixed transmission power for all users
 - c) Increasing frequency bandwidth
 - d) Eliminating soft handoffs
23. Which multiple access technique divides the frequency band into separate channels?
- a) TDMA

- b) **FDMA**
- c) CDMA
- d) OFDMA

24. In TDMA, how do multiple users share the same frequency channel?
- a) By using unique spreading codes
 - b) **By dividing time into sequential slots**
 - c) By varying transmission power
 - d) By frequency hopping
25. Which technology uses orthogonal codes to separate users in the same frequency band?
- a) FDMA
 - b) TDMA
 - c) **CDMA**
 - d) SDMA
26. What is a key advantage of CDMA over TDMA/FDMA?
- a) Simpler hardware design
 - b) **Better resistance to interference and higher capacity**
 - c) Fixed time slot allocation
 - d) No need for power control
27. Which generation of cellular networks primarily used FDMA?
- a) 2G (GSM)
 - b) **1G (Analog systems like AMPS)**
 - c) 3G (UMTS)
 - d) 4G (LTE)
28. What is a disadvantage of FDMA compared to TDMA?
- a) Complex synchronization
 - b) **Inefficient spectrum usage (guard bands required)**
 - c) Vulnerable to multipath fading
 - d) Limited scalability
29. Which technology combines TDMA and FDMA?
- a) CDMA
 - b) **GSM**
 - c) WCDMA
 - d) OFDMA
30. Why does CDMA require precise power control?
- a) To minimize battery usage
 - b) **To avoid the "near-far problem"**
 - c) To reduce handover frequency
 - d) To simplify base station design

SECTION B

Comprehensive Answer Questions

Attempt any five (5) questions out of eight (8) questions

[5 x 6 = 30]

1. Define "cell breathing" in CDMA networks and explain its root cause using the near-far problem concept. [Chapter 7: Intelligent Cell Concept and Application] (6 marks)
2. Discuss the role of intelligent cells in enabling smart city functionalities, with concrete examples demonstrating their applications. [Chapter 7: Intelligent Cell Concept and Application] (6 marks)
3. How does cell sectorization modify the effective cluster size and interference pattern? [Chapter 4: Cellular System Design Fundamentals] (6 marks)
4. Compare BRI (Basic Rate Interface) and PRI (Primary Rate Interface) with respect to:
 - a. Channel structure (B/D channels)
 - b. Transmission rate
 - c. Typical use cases[Chapter 6: Wireless Networking] (6 marks)
5. Define the term "wireless generation" and list the key technological advancements introduced in each generation (1G to 5G). [Chapter 1: Introduction to Wireless Communication] (6 marks)
6. What are the technical advantages and limitations of DECT technology in modern wireless communication systems, and in what applications does DECT remain a viable solution compared to alternatives like Wi-Fi, Bluetooth, and VoIP? [Chapter 1: Introduction to Wireless Communication] (6 marks)
7. A cellular service provider observes that in a particular cell, subscribers make:
 - a. 300 calls in 1 hour
 - b. Average call duration = 2 minutesCalculate: Traffic load in Erlangs.
[Chapter 4: Cellular System Design Fundamentals] (6 marks)
8. Short question answer:
 - a. Fixed Telephone Network
 - b. Fixed Channel Assignments[Chapter 1 & Chapter 6] (6 marks)

SECTION C

Long Answer Questions

Attempt any two (2) questions out of three (3) questions.

[2 x 20 = 40]

1. a. Design and explain a wireless network infrastructure to provide basic 4G/LTE internet and voice communication services to the majority of the population in Jumla District. Consider topographical, economic, technological, and social constraints. [Chapter 2: Modern Wireless Communication System] (15 marks)
b. A cellular network provider observes poor Quality of Service (QoS) in densely populated areas. Suggest three technical strategies to improve QoS, linking each strategy to specific performance criteria. [Chapter 3: Introduction to Cellular System] (5 marks)
(20 marks) (15+5)
2. a. A GSM network operates in the 900 MHz band with the following specifications:
 - Total available bandwidth = 25 MHz
 - Carrier spacing (channel bandwidth) = 200 kHz
 - TDMA frames per carrier = 8 time slots
 - Frequency reuse factor (K) = 4 (assume 4-cell reuse pattern)Calculate:
 - i. The total number of RF carriers (channels) available in the system.
 - ii. The number of traffic channels per cell.
 - iii. If the system uses a 12-cell cluster (instead of 4), how does it affect the number of traffic channels per cell?[Chapter 3: Introduction to Cellular System] (15 marks)
- b. An FDMA-based cellular system has the following specifications:
 - Total available bandwidth = 10 MHz
 - Each voice channel requires = 30 kHz
 - Guard band between channels = 10 kHzCalculate:
 - i. Total number of channels (including guard bands).
 - ii. Net bandwidth used by voice channels (without guard bands).[Chapter 5: Multiple Access Techniques for Wireless Communication] (5 marks)
(20 marks) (15+5)

3. a. Critically analyze the design and performance of the following random-access protocols in wireless networks:

i. ALOHA (Pure and Slotted)

ii. CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance)

iii. CSMA/CD (Carrier Sense Multiple Access with Collision Detection)

[Chapter 5: Multiple Access Techniques for Wireless Communication] (15 marks)

b. Define SS7 and explain its role in modern telecommunication networks. [Chapter 6: Wireless Networking] (5 marks)

(20 marks)